

7.2 EFFICIENCY & SOCIAL SURPLUS

Microeconomics by Topic
7. Demand & Supply (II)

MARKING SCHEME

1985/AL/I/22 D	1997/AL/I/07 C	2003/AL/I/11 A	2012/AL/I/05 B (45%)	2012/DSE/I/14 C (64%)
1986/AL/I/15 C	1998/AL/I/12 D	2006/AL/I/02 B (71%)	2012/AL/I/07 B (68%)	2014/DSE/I/15 C (54%)
1990/AL/I/02 A	1998/AL/I/24 A	2006/AL/I/23 C (53%)	2012/AL/I/16 D (67%)	2015/DSE/I/22 D (64%)
1991/AL/I/08 D	2000/AL/I/10 A	2008/AL/I/05 C (94%)	PP/DSE/I/12 B	2016/DSE/I/21 C (68%)
1995/AL/I/03 B	2001/AL/I/19 B	2010/AL/I/07 A (85%)	PP/DSE/I/14 A	2016/DSE/I/22 D (61%)
1995/AL/I/06 B	2002/AL/I/10 D	2012/AL/I/01 B (70%)	PP/DSE/I/17 C	2017/DSE/I/12 A (88%)
2019/DSE/I/21 B	2019/DSE/I/22 D	2021/DSE/I/17 A	2021/DSE/I/19 C	2021/DSE/I/20 A
2022/DSE/I/10 C	2022/DSE/I/20 D	2023/DSE/I/20 D	2023/DSE/I/21 C	2023/DSE/I/22 D
2024/DSE/I/19 B	2024/DSE/I/20 C	2024/DSE/I/21 C	2024/DSE/I/22 D	2025/DSE/I/12 B
2025/DSE/I/18 B	2025/DSE/I/21 D	2025/DSE/I/22 B	2025/DSE/I/43 C	

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1988/AL/I/2

The marginal benefit (i.e. value) is the maximum a consumer is willing to pay for the last unit purchased. Errors aside, no maximizing individual will, on a voluntary basis, pay more than what he is willing to pay.

If price is lower than his marginal benefit, to maximise consumer surplus, the consumer will buy more until the price is equal to his marginal benefit. Therefore, at the margin, he will pay the maximum of what he is willing to pay, and price equals marginal benefit.

1989/AL/I/1

Price is a marginal benefit (i.e. the maximum an individual is willing to pay at the margin).

Price serves as a signaling device to direct resource allocation (i.e. allocative function); it also serves as a criterion of determining who wins or loses in competing for goods and services in the market place (i.e. rationing function).

1992/AL/I/6

The market price of the permits will definitely be higher than \$20, because in this market the criteria of drawing lottery will not be used, leaving the market price to measure fully the marginal benefit people place on the permit.

With the permits now going to those who offer the highest prices in the resale market, more goats will likely be killed. This is because the higher permit price implies that either the permit holders are better, more enthusiastic, or more frequent hunters who place a higher marginal benefit on hunting.

(Note: More goats would be killed not because permit holders wished to make compensation for the higher price, but because only better hunters were willing to pay a higher price to obtain the permits.)

1994/AL/I/8

No. The marginal cost of reproducing discs (legal or illegal) is much lower than the selling price. Since illegal copies are sold at a lower price, buyers can enjoy a greater consumer surplus. The marginal benefit is closer to the marginal cost, leading to a more efficient allocation of resources.

Yes. If reproduced discs are sold in the market, in an extreme case, the composers of the songs will not be able to receive any revenue from selling the original discs. There will be no more new songs and original discs in the future (and hence illegal reproduction), and no more consumer surplus and producer surplus in the future.

1997/AL/I/8(a)

When a student is required to pay a high tuition out of his own pocket to buy university education, his own marginal benefit on university education will also be high. He will therefore treasure education more, with the result that he would study harder than when he does not have to pay and put a low marginal benefit on education.

(Note: Whether students are willing to work harder or not depends on the price they have to pay for being lazy, as indicated by the law of demand. The 'price' here is the value of learning forgone.)

2002/AL/I/06(b)

Whoever pays for the purchase of the tunnel, the payment is a cost to society. If the government purchases the tunnel (tax money is a cost to society) and let all cars use it free of charge, there is no way of knowing the total benefit to the users will cover the purchasing cost of the tunnel, even if we ignore the cost of operation. Whether or not there is congestion makes no difference to this answer.

2006/AL/I/5

In a buffet dinner when a consumer eats until his marginal benefit reaches zero, lower than the marginal cost of producing food, we have an inefficient situation.

2006/AL/I/9

It does not mean ladies in mainland China are richer than those in Hong Kong. It only indicates that mainland ladies have a higher maximum willingness to pay for prestigious handbags. Such a higher marginal benefit could be a result of higher wealth or income, or other factors such as a lack of substitutes or consumer preference (e.g. due to culture or advertisement) favourable to these handbags.

Many ladies from China pay such high prices because they do care for famous or prestigious brand names and it is possible that mainland ladies in general have a higher preference for prestige.

2009/AL/I/7(a)

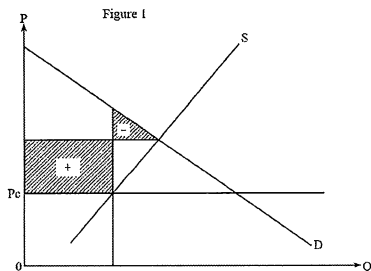
The higher licence fees plus the higher travelling costs indicate that the cost per animal hunted is considerably higher for out-of-state hunters, so in general they would have to be better hunters to equate the marginal gain and marginal cost of hunting.

PP/DSE/I/4

(a) Deadweight loss is the loss in the total social surplus or the potential gains not fully captured when marginal benefit does not equal marginal cost. (2)

(b) (i) Yes.

(1)



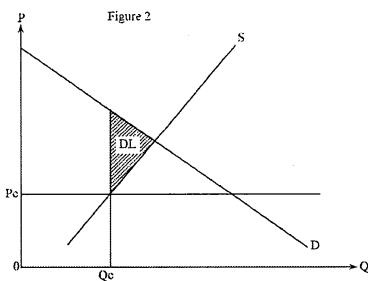
Indicate in the diagram:

- gain of consumer surplus (+)
- loss of consumer surplus (-)

(1)

(1)

(b) (ii)



Indicate in the diagram:

- deadweight loss (DL)
- quantity is Q_c

(1)

(1)

Verbal elaboration:

No, because

when price is set below equilibrium, the quantity transacted will be reduced.

marginal cost of sellers is lower than marginal benefit of consumers / there is deadweight loss.

(1)

(1)

(1)

PP/DSE/I/12(b)(i)

Rationing function: Tickets were distributed to the buyers who were willing to pay at least \$200.

(1)

2015/DSE/I/4(a)

Allocative function: Higher prices of small-sized flats would induce real-estate developers to direct more resources towards the production of small flats.

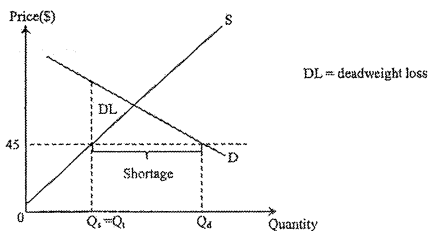
OR

Signal function: Prices help transmit information about consumer demand relative to producer supply.

(2)

2016/DSE/II/11

(b)



Illustrate in the diagram:

- price set below equilibrium (1)
- correct position of Q_s (1)
- correct position of deadweight loss (1)

Verbal elaboration:

The price is set at \$45 which is below equilibrium. (1)

(With the existence of shortage,) the quantity transacted is below equilibrium quantity or there is underproduction. (1)

the marginal benefit is larger than the marginal cost. (1)

(d) Efficiency:

The fee increase would improve efficiency as it results in a price closer to the equilibrium level, thus narrowing the gap between MB and MC and reducing the size of deadweight loss. (2)

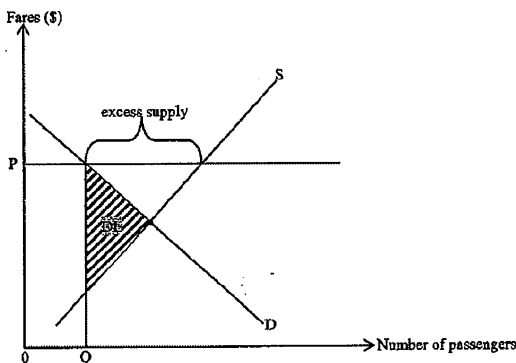
2019/DSE/II/11

- (a) The launch of HSR will increase the geographical mobility of workers because the transportation network improves. (3)

- (b) Verbal elaboration: (2)
Fares are set above the market-clearing (or equilibrium) level. So marginal benefit of train service is larger than marginal cost.

Illustrate in the diagram:

- correct position of P and Q (1)
- correct position of deadweight loss (DL) (1)



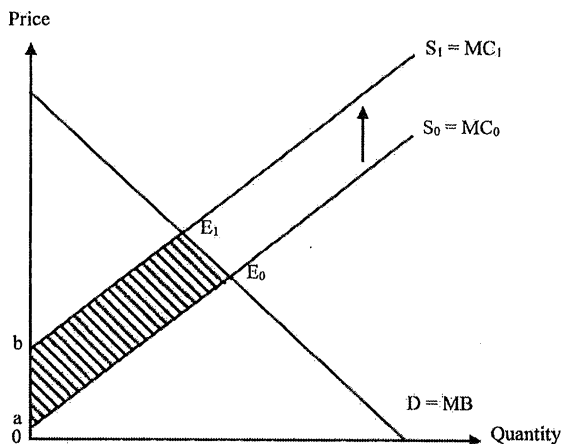
2023/DSE/II/5a-b

(a) Indicate in the diagram:

- upward shift of MC curve (leftward shift of supply curve)
- decrease in total social surplus

(1)

(1)



Decrease in total social surplus = area (abE₁E₀)

(b) Verbal elaboration:

No, efficiency is still attained because at the new equilibrium quantity, marginal benefit is still equal to marginal cost.

(1)

(1)

2023/DSE/II/11b

(b) rationing function: concert tickets are distributed to the highest-valued buyers.

(2)

2025/DSE/II/8

(a) (i) Indicate in the diagram:

- correct position of P_{W+t}
- correct position of Q_1

(1)

(1)

(ii) Indicate in the diagram:

- correct position of Q_M

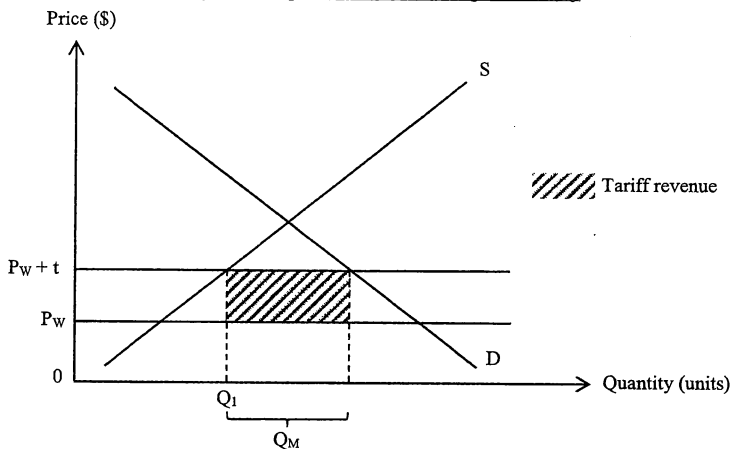
(1)

(iii) Indicate in the diagram:

- correct position of tariff revenue

(1)

Figure 1: Market for agricultural products in the small open economy



- (b) As import volume decreases while the world price remains unchanged, the total import value of agricultural products in the economy decreases. Total import value of goods decreases. Trade balance increases. (3)